Documentation of CCNA final project by Omer Asraf

Introduction

This project focuses on the design, implementation, and configuration of an enterprise network infrastructure for AI-TECH, a company operating two geographically separated sites: Afek and Tel Aviv. The network was implemented using Cisco Packet Tracer and follows enterprise networking best practices, including hierarchical design, VLAN segmentation, dynamic routing, centralized services, and security controls.

Background

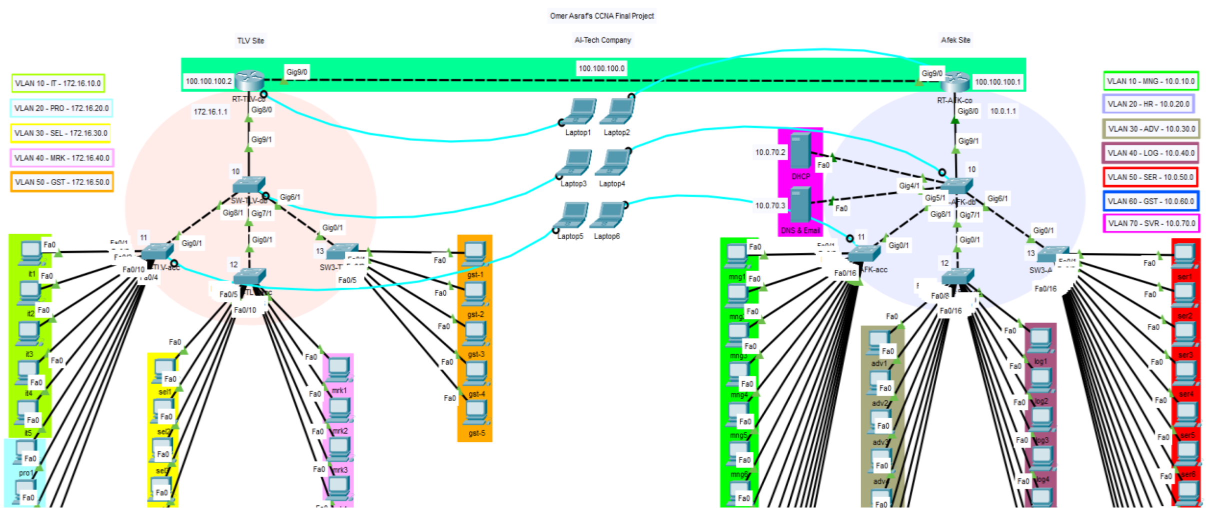
AI-TECH operates multiple organizational departments across its two sites, each with different operational and security requirements. To support these needs, a scalable and secure network architecture was required, providing logical separation between departments while maintaining controlled connectivity and centralized service availability. The network design follows a hierarchical model consisting of Access, Distribution, and Core layers, with inter-site connectivity achieved through a dedicated WAN link and dynamic routing.

Goals

The goals of this project are to design and deploy a multi-site enterprise network, implement VLAN-based segmentation, enable inter-VLAN routing, establish secure WAN connectivity, deploy centralized network services, apply dynamic and static IP addressing, enforce security policies using ACLs, and ensure reliable, secure, and manageable network operations in accordance with CCNA-level enterprise standards.

To view and download the project file (pkt) on Github **click here**

Network Topology



Documentation

Note: In the project documentation that you will see before you, screenshots will sometimes appear that show commands and settings that were performed on only one of the communication components and/or end users - this is for example and illustration purposes.

1. The network was designed and implemented using a bottom-up approach, starting with the Access layer to define user-to-VLAN associations, followed by the Distribution layer for aggregation and Trunking, and finally the Core layer to enable routing between the established networks.

```
SW1-AFK-acc#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#vlan 10
SW1-AFK-acc(config-vlan)#na mng
SW1-AFK-acc(config-vlan)#ex
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#int ra fa0/1-8
SW1-AFK-acc(config-if-range)#sw m a
SW1-AFK-acc(config-if-range)#sw a vl 10
SW1-AFK-acc(config-if-range)#ex
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#int ra fa0/9-16
SW1-AFK-acc(config-if-range)#sw m a
SW1-AFK-acc(config-if-range)#sw a vl 20
SW1-AFK-acc(config-if-range)#ex
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#end
SW1-AFK-acc#
%SYS-5-CONFIG_I: Configured from console by console

SW1-AFK-acc#
SW1-AFK-acc#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   mng                              active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
20   hr                               active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1-AFK-acc#
```

```
SW1-TLV-acc>en
SW1-TLV-acc#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#vlan 10
SW1-TLV-acc(config-vlan)#na it
SW1-TLV-acc(config-vlan)#ex
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#vlan 20
SW1-TLV-acc(config-vlan)#na pro
SW1-TLV-acc(config-vlan)#ex
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#int ra fa0/1-5
SW1-TLV-acc(config-if-range)#sw m a
SW1-TLV-acc(config-if-range)#sw a vl 10
SW1-TLV-acc(config-if-range)#ex
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#int ra fa0/6-10
SW1-TLV-acc(config-if-range)#sw m a
SW1-TLV-acc(config-if-range)#sw a vl 20
SW1-TLV-acc(config-if-range)#ex
SW1-TLV-acc(config)#end
SW1-TLV-acc#
%SYS-5-CONFIG_I: Configured from console by console

SW1-TLV-acc#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   it                               active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5
20   pro                              active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1-TLV-acc#
```

2. At the Access Layer, Port Security was implemented to restrict each access port to a single device using Sticky MAC addressing and a restrict violation mode, ensuring that unauthorized devices are blocked while keeping the port operational.

```
SW1-AFK-acc>en
SW1-AFK-acc#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#int ra fa0/1-16
SW1-AFK-acc(config-if-range)#sw m a
SW1-AFK-acc(config-if-range)#sw po
SW1-AFK-acc(config-if-range)#sw po max 1
SW1-AFK-acc(config-if-range)#sw po mac s
SW1-AFK-acc(config-if-range)#sw po v r
SW1-AFK-acc(config-if-range)#ex
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#int ra fa0/17-24
SW1-AFK-acc(config-if-range)#sw m a
SW1-AFK-acc(config-if-range)#sw po
SW1-AFK-acc(config-if-range)#sw po v sh
SW1-AFK-acc(config-if-range)#ex
SW1-AFK-acc(config)#end
SW1-AFK-acc#
%SYS-5-CONFIG_I: Configured from console by console

SW1-AFK-acc#sh p
SW1-AFK-acc#sh port
SW1-AFK-acc#sh port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)          (Count)
--------------------------------------------------------------------
        Fa0/1        1          0                  0         Restrict
        Fa0/2        1          0                  0         Restrict
        Fa0/3        1          0                  0         Restrict
        Fa0/4        1          0                  0         Restrict
        Fa0/5        1          0                  0         Restrict
        Fa0/6        1          0                  0         Restrict
        Fa0/7        1          0                  0         Restrict
        Fa0/8        1          0                  0         Restrict
        Fa0/9        1          0                  0         Restrict
       Fa0/10        1          0                  0         Restrict
       Fa0/11        1          0                  0         Restrict
       Fa0/12        1          0                  0         Restrict
       Fa0/13        1          0                  0         Restrict
       Fa0/14        1          0                  0         Restrict
       Fa0/15        1          0                  0         Restrict
       Fa0/16        1          0                  0         Restrict
       Fa0/17        1          0                  0         Shutdown
       Fa0/18        1          0                  0         Shutdown
       Fa0/19        1          0                  0         Shutdown
       Fa0/20        1          0                  0         Shutdown
       Fa0/21        1          0                  0         Shutdown
       Fa0/22        1          0                  0         Shutdown
       Fa0/23        1          0                  0         Shutdown
       Fa0/24        1          0                  0         Shutdown
----------------------------------------------------------------------
```

```
SW1-TLV-acc>en
SW1-TLV-acc#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#int ra fa0/1-10
SW1-TLV-acc(config-if-range)#sw m a
SW1-TLV-acc(config-if-range)#sw po
SW1-TLV-acc(config-if-range)#sw po max 1
SW1-TLV-acc(config-if-range)#sw po mac s
SW1-TLV-acc(config-if-range)#sw po v r
SW1-TLV-acc(config-if-range)#ex
SW1-TLV-acc(config)#
SW1-TLV-acc(config)#int ra fa0/11-24
SW1-TLV-acc(config-if-range)#sw m a
SW1-TLV-acc(config-if-range)#sw po
SW1-TLV-acc(config-if-range)#sw po v sh
SW1-TLV-acc(config-if-range)#ex
SW1-TLV-acc(config)#end
SW1-TLV-acc#
%SYS-5-CONFIG_I: Configured from console by console

SW1-TLV-acc#sh port
SW1-TLV-acc#sh port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Actio
               (Count)       (Count)          (Count)
--------------------------------------------------------------------
        Fa0/1        1          0                  0         Restrict
        Fa0/2        1          0                  0         Restrict
        Fa0/3        1          0                  0         Restrict
        Fa0/4        1          0                  0         Restrict
        Fa0/5        1          0                  0         Restrict
        Fa0/6        1          0                  0         Restrict
        Fa0/7        1          0                  0         Restrict
        Fa0/8        1          0                  0         Restrict
        Fa0/9        1          0                  0         Restrict
       Fa0/10        1          0                  0         Restrict
       Fa0/11        1          0                  0         Shutdown
       Fa0/12        1          0                  0         Shutdown
       Fa0/13        1          0                  0         Shutdown
       Fa0/14        1          0                  0         Shutdown
       Fa0/15        1          0                  0         Shutdown
       Fa0/16        1          0                  0         Shutdown
       Fa0/17        1          0                  0         Shutdown
       Fa0/18        1          0                  0         Shutdown
       Fa0/19        1          0                  0         Shutdown
       Fa0/20        1          0                  0         Shutdown
       Fa0/21        1          0                  0         Shutdown
       Fa0/22        1          0                  0         Shutdown
       Fa0/23        1          0                  0         Shutdown
       Fa0/24        1          0                  0         Shutdown
----------------------------------------------------------------------
```

3. At the Distribution Layer, all VLANs were centralized and aggregated, with dedicated trunk links configured toward each Access switch and a central trunk toward the Core router, without performing any routing at this layer.

```
SW-AFK-db>en
SW-AFK-db#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 10
SW-AFK-db(config-vlan)#na mng
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 20
SW-AFK-db(config-vlan)#na hr
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 30
SW-AFK-db(config-vlan)#na adv
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 40
SW-AFK-db(config-vlan)#na log
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 50
SW-AFK-db(config-vlan)#na ser
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 60
SW-AFK-db(config-vlan)#na gst
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#vlan 70
SW-AFK-db(config-vlan)#na svr
SW-AFK-db(config-vlan)#ex
SW-AFK-db(config)#end
SW-AFK-db#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW-AFK-db#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 8/1
SW-AFK-db(config-if)#sw m t

SW-AFK-db(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/1, changed state to up

SW-AFK-db(config-if)#sw t al
SW-AFK-db(config-if)#sw t allowed vlan 10,20
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 7/1
SW-AFK-db(config-if)#sw m t

SW-AFK-db(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet7/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet7/1, changed state to up

SW-AFK-db(config-if)#sw t allowed vlan 30,40
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 6/1
SW-AFK-db(config-if)#sw m t

SW-AFK-db(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet6/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet6/1, changed state to up

SW-AFK-db(config-if)#sw t allowed vlan 50,60
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 5/1
SW-AFK-db(config-if)#sw m t

SW-AFK-db(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet5/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet5/1, changed state to up

SW-AFK-db(config-if)#sw t allowed vlan 70
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 4/1
SW-AFK-db(config-if)#sw m t

SW-AFK-db(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet4/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet4/1, changed state to up

SW-AFK-db(config-if)#sw t allowed vlan 70
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#end
SW-AFK-db#
```

```
SW-AFK-db#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------
1    default                          active    Gig9/1
10   mng                              active
20   hr                               active
30   adv                              active
40   log                              active
50   ser                              active
60   gst                              active
70   svr                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-AFK-db#
SW-AFK-db#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Gig4/1      on           802.1q         trunking      1
Gig5/1      on           802.1q         trunking      1
Gig6/1      on           802.1q         trunking      1
Gig7/1      on           802.1q         trunking      1
Gig8/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig4/1      70
Gig5/1      70
Gig6/1      50,60
Gig7/1      30,40
Gig8/1      10,20

Port        Vlans allowed and active in management domain
Gig4/1      70
Gig5/1      70
Gig6/1      50,60
Gig7/1      30,40
Gig8/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig4/1      70
Gig5/1      70
Gig6/1      50,60
Gig7/1      30,40
Gig8/1      10,20
```

```
SW-AFK-db(config)#int gig 4/1
SW-AFK-db(config-if)#sw m a
SW-AFK-db(config-if)#sw a vl 70
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 5/1
SW-AFK-db(config-if)#sw m a
SW-AFK-db(config-if)#sw a vl 70
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#end
SW-AFK-db#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW-AFK-db#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW-AFK-db(config)#
SW-AFK-db(config)#int gig 9/1
SW-AFK-db(config-if)#sw m t
SW-AFK-db(config-if)#sw t allowed vlan 10,20,30,40,50,60,70
SW-AFK-db(config-if)#end
SW-AFK-db#
```

```
SW-TLV-db#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig9/1
10   it                               active
20   pro                              active
30   sel                              active
40   mrk                              active
50   gst                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-TLV-db#
SW-TLV-db#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Gig6/1      on           802.1q         trunking      1
Gig7/1      on           802.1q         trunking      1
Gig8/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig6/1      50
Gig7/1      30,40
Gig8/1      10,20

Port        Vlans allowed and active in management domain
Gig6/1      50
Gig7/1      30,40
Gig8/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig6/1      50
Gig7/1      30,40
Gig8/1      10,20
```

Section 4 in next page

4. At the Core Layer, a Router-on-a-Stick architecture was implemented, where each VLAN was assigned a dedicated sub-interface with IEEE 802.1Q encapsulation and a unique default gateway, enabling full inter-VLAN routing across all organizational departments.

```
RT-AFK-co>en
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.10
RT-AFK-co(config-subif)#en d 10
RT-AFK-co(config-subif)#ip add 10.0.10.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.20
RT-AFK-co(config-subif)#en d 20
RT-AFK-co(config-subif)#ip add 10.0.20.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.30
RT-AFK-co(config-subif)#en d 30
RT-AFK-co(config-subif)#ip add 10.0.30.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.40
RT-AFK-co(config-subif)#en d 40
RT-AFK-co(config-subif)#ip add 10.0.40.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.50
RT-AFK-co(config-subif)#en d 50
RT-AFK-co(config-subif)#ip add 10.0.50.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.60
RT-AFK-co(config-subif)#en d 60
RT-AFK-co(config-subif)#ip add 10.0.60.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.70
RT-AFK-co(config-subif)#ip add 10.0.70.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

RT-AFK-co(config-subif)#en d 70
RT-AFK-co(config-subif)#ip add 10.0.70.1 255.255.255.0
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#end
```

```
RT-AFK-co(config)#int gig 8/0
RT-AFK-co(config-if)#no sh

RT-AFK-co(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet8/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.30, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.40, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.50, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.60, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.70, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.70, changed state to up
```

```
RT-AFK-co#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet8/0     unassigned      YES unset  up                    up
GigabitEthernet8/0.10  10.0.10.1       YES manual up                    up
GigabitEthernet8/0.20  10.0.20.1       YES manual up                    up
GigabitEthernet8/0.30  10.0.30.1       YES manual up                    up
GigabitEthernet8/0.40  10.0.40.1       YES manual up                    up
GigabitEthernet8/0.50  10.0.50.1       YES manual up                    up
GigabitEthernet8/0.60  10.0.60.1       YES manual up                    up
GigabitEthernet8/0.70  10.0.70.1       YES manual up                    up
GigabitEthernet9/0     unassigned      YES unset  administratively down down
RT-AFK-co#
```

mng1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::203:E4FF:FE6B:7110
   IPv6 Address....................: ::
   IPv4 Address....................: 10.0.10.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     10.0.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 10.0.50.2

Pinging 10.0.50.2 with 32 bytes of data:

Reply from 10.0.50.2: bytes=32 time<1ms TTL=127
Reply from 10.0.50.2: bytes=32 time<1ms TTL=127
Reply from 10.0.50.2: bytes=32 time<1ms TTL=127
Reply from 10.0.50.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
RT-TLV-co>en
RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.10
RT-TLV-co(config-subif)#en d 10
RT-TLV-co(config-subif)#ip add 172.16.10.1 255.255.255.0
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.20
RT-TLV-co(config-subif)#en d 20
RT-TLV-co(config-subif)#ip add 172.16.20.1 255.255.255.0
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.30
RT-TLV-co(config-subif)#en d 30
RT-TLV-co(config-subif)#ip add 172.16.30.1 255.255.255.0
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.40
RT-TLV-co(config-subif)#en d 40
RT-TLV-co(config-subif)#ip add 172.16.40.1 255.255.255.0
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.50
RT-TLV-co(config-subif)#en d 50
RT-TLV-co(config-subif)#ip add 172.16.50.1 255.255.255.0
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0
RT-TLV-co(config-if)#no sh

RT-TLV-co(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet8/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.30, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.40, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet8/0.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.50, changed state to up
```

```
RT-TLV-co#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet8/0     unassigned      YES unset  up                    up
GigabitEthernet8/0.10  172.16.10.1     YES manual up                    up
GigabitEthernet8/0.20  172.16.20.1     YES manual up                    up
GigabitEthernet8/0.30  172.16.30.1     YES manual up                    up
GigabitEthernet8/0.40  172.16.40.1     YES manual up                    up
GigabitEthernet8/0.50  172.16.50.1     YES manual up                    up
GigabitEthernet9/0     unassigned      YES unset  administratively down down
RT-TLV-co#
```

5. To secure network devices, an encrypted privileged access password was configured using enable secret, along with a console password, and service password-encryption was enabled to prevent plaintext password exposure in configuration files.

```
RT-AFK-co>en
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#enable secret admin123
RT-AFK-co(config)#service password-encryption
RT-AFK-co(config)#line console 0
RT-AFK-co(config-line)#password console123
RT-AFK-co(config-line)#login
RT-AFK-co(config-line)#ex
% Ambiguous command: "ex"
RT-AFK-co(config-line)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-AFK-co#ex

RT-AFK-co con0 is now available

Press RETURN to get started.

User Access Verification

Password:

RT-AFK-co>en
Password:
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
```

6. Secure remote management was configured using SSH only, with local user authentication and RSA encryption, while insecure legacy access methods such as Telnet were explicitly disabled.

```
User Access Verification

Password:

RT-TLV-co>en
Password:
RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip domain-name ai-tech.local
RT-TLV-co(config)#username admin privilege 15 secret sshadmin123
RT-TLV-co(config)#crypto key generate rsa
The name for the keys will be: RT-TLV-co.ai-tech.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

RT-TLV-co(config)#ip ssh version 2
*Mar 1 5:9:27.854: %SSH-5-ENABLED: SSH 1.99 has been enabled
RT-TLV-co(config)#line vty 0 15
RT-TLV-co(config-line)#login local
RT-TLV-co(config-line)#transport input ssh
RT-TLV-co(config-line)#exec-timeout 10 0
RT-TLV-co(config-line)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-TLV-co#sh ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
RT-TLV-co#
```

7. A dedicated Management Network was implemented to isolate infrastructure management traffic from user traffic, based on a management VLAN and the addressing schemes 10.0.1.0/24 and 172.16.1.0/24 for the Afek and Tel Aviv sites respectively.

```
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.111
RT-AFK-co(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet8/0.111, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet8/0.111, changed state to
up

RT-AFK-co(config-subif)#en d 111
RT-AFK-co(config-subif)#ip add 10.0.1.1 255.255.255.0
RT-AFK-co(config-subif)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console
```

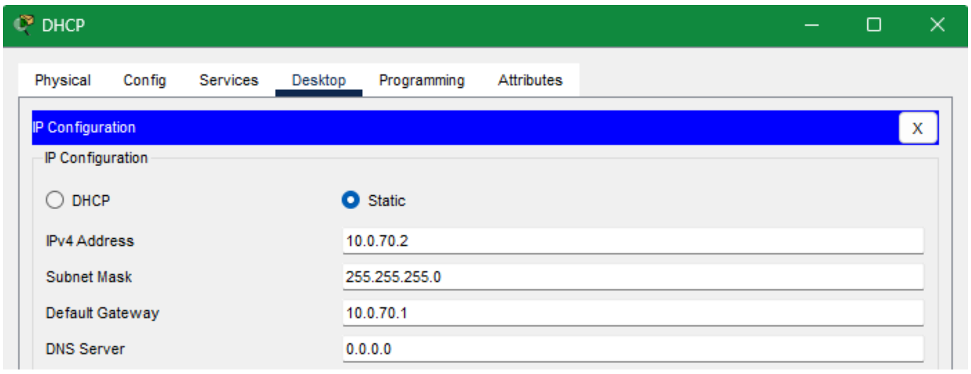
```
SW-AFK-db#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW-AFK-db(config)#
SW-AFK-db(config)#int vlan 111
SW-AFK-db(config-if)#
%LINK-5-CHANGED: Interface Vlan111, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan111, changed state to up

SW-AFK-db(config-if)#ip add 10.0.1.10 255.255.255.0
SW-AFK-db(config-if)#no sh
SW-AFK-db(config-if)#ex
SW-AFK-db(config)#ip default-gateway 10.0.1.1
SW-AFK-db(config)#end
SW-AFK-db#
%SYS-5-CONFIG_I: Configured from console by console

SW-AFK-db#Enter configuration commands, one per line.  End with CNTL/Z.
SW-AFK-db(config)#
```

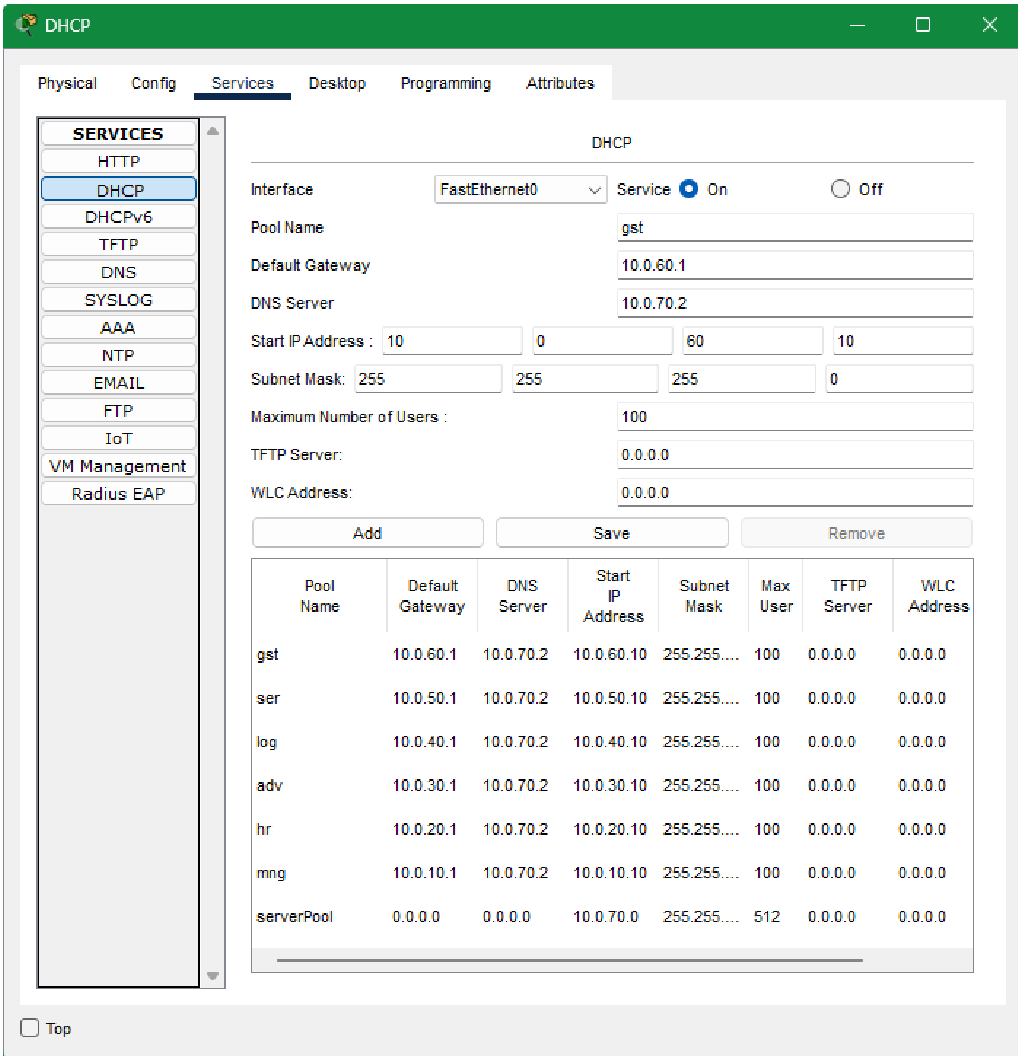
```
SW1-AFK-acc#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
SW1-AFK-acc(config)#
SW1-AFK-acc(config)#int vlan 111
SW1-AFK-acc(config-if)#
%LINK-5-CHANGED: Interface Vlan111, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan111, changed state to up

SW1-AFK-acc(config-if)#ip add 10.0.1.11 255.255.255.0
SW1-AFK-acc(config-if)#no sh
SW1-AFK-acc(config-if)#ex
SW1-AFK-acc(config)#ip default-gateway 10.0.1.1
SW1-AFK-acc(config)#end
SW1-AFK-acc#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW-AFK-db#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   mng                              active
20   hr                               active
30   adv                              active
40   log                              active
50   ser                              active
60   gst                              active
70   svr                              active    Gig4/1, Gig5/1
111  net                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-AFK-db#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Gig6/1      on           802.1q         trunking      1
Gig7/1      on           802.1q         trunking      1
Gig8/1      on           802.1q         trunking      1
Gig9/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig6/1      50,60,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,60,70,111

Port        Vlans allowed and active in management domain
Gig6/1      50,60,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,60,70,111

Port        Vlans in spanning tree forwarding state and not pruned
Gig6/1      50,60,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,60,70,111

SW-AFK-db#
SW-AFK-db#
SW-AFK-db#
SW-AFK-db#
SW-AFK-db#
SW-AFK-db#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet4/1     unassigned      YES manual up                    up
GigabitEthernet5/1     unassigned      YES manual up                    up
GigabitEthernet6/1     unassigned      YES manual up                    up
GigabitEthernet7/1     unassigned      YES manual up                    up
GigabitEthernet8/1     unassigned      YES manual up                    up
GigabitEthernet9/1     unassigned      YES manual up                    up
Vlan1                  unassigned      YES manual administratively down down
Vlan111                10.0.1.10       YES manual up                    up
```

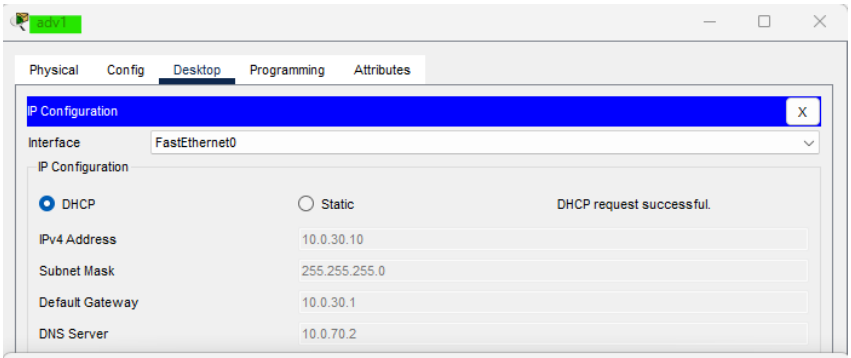
```
SW-TLV-db#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   it                               active
20   pro                              active
30   sel                              active
40   mrk                              active
50   gst                              active
111  net                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW-TLV-db#
SW-TLV-db#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet6/1     unassigned      YES manual up                    up
GigabitEthernet7/1     unassigned      YES manual up                    up
GigabitEthernet8/1     unassigned      YES manual up                    up
GigabitEthernet9/1     unassigned      YES manual up                    up
Vlan1                  unassigned      YES manual administratively down down
Vlan111                172.16.1.10     YES manual up                    up
SW-TLV-db#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Gig6/1      on           802.1q         trunking      1
Gig7/1      on           802.1q         trunking      1
Gig8/1      on           802.1q         trunking      1
Gig9/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig6/1      50,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,111

Port        Vlans allowed and active in management domain
Gig6/1      50,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,111

Port        Vlans in spanning tree forwarding state and not pruned
Gig6/1      50,111
Gig7/1      30,40,111
Gig8/1      10,20,111
Gig9/1      10,20,30,40,50,111
```

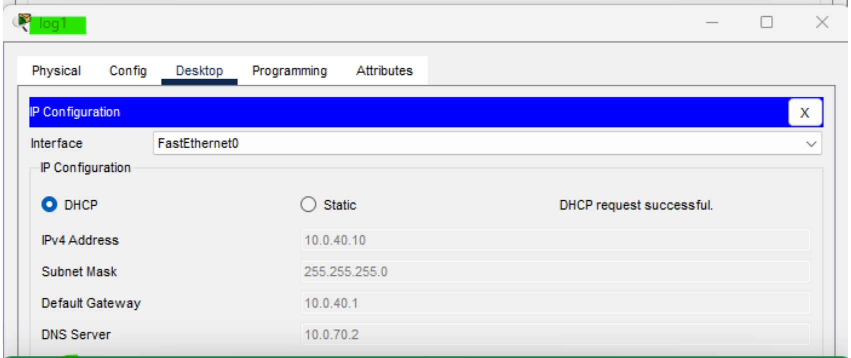
8. A centralized DHCP Server (10.0.70.2) located at the Afek site provides dynamic IP address allocation for all user VLANs at that site using DHCP Relay (ip helper-address) configured on the router interfaces.

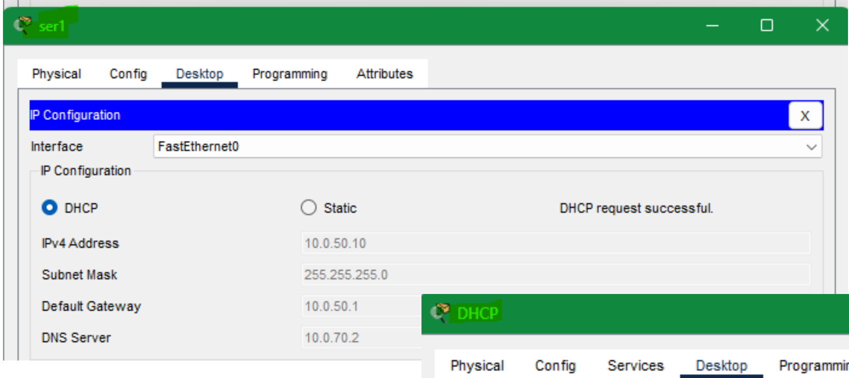


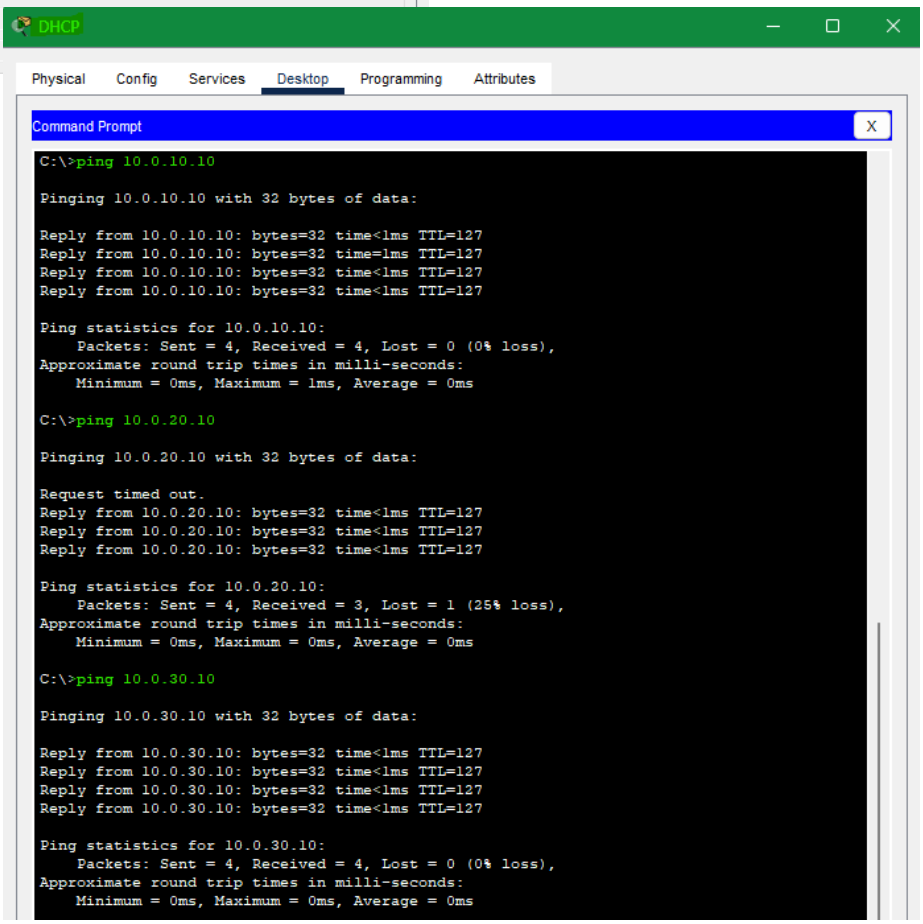



```
RT-AFK-co>en
Password:
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.10
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.20
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.30
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.40
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.50
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.60
RT-AFK-co(config-subif)#ip helper-address 10.0.70.2
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console
```


adv1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
10.0.30.10
Subnet Mask
255.255.255.0
Default Gateway
10.0.30.1
DNS Server
10.0.70.2


log1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
10.0.40.10
Subnet Mask
255.255.255.0
Default Gateway
10.0.40.1
DNS Server
10.0.70.2


ser1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
10.0.50.10
Subnet Mask
255.255.255.0
Default Gateway
10.0.50.1
DNS Server
10.0.70.2

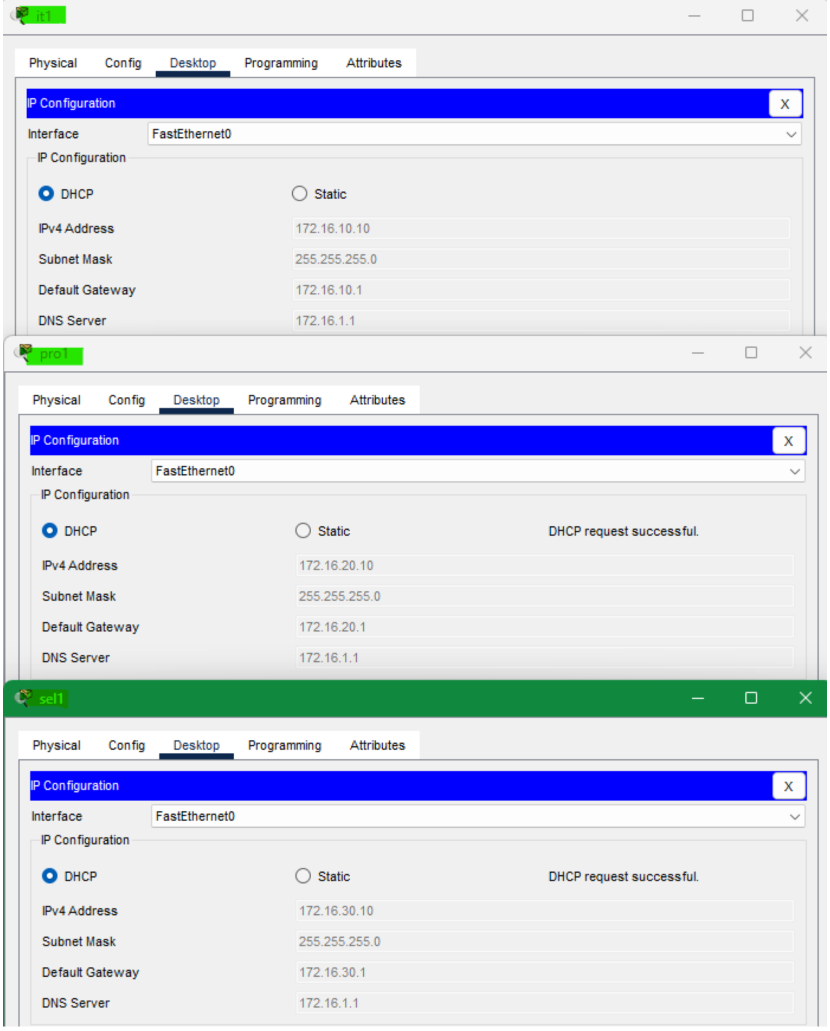

DHCP
Physical
Config
Services
Desktop
Programming
Attributes
Command Prompt
X
C:\>ping 10.0.10.10
Pinging 10.0.10.10 with 32 bytes of data:
Reply from 10.0.10.10: bytes=32 time<1ms TTL=127
Reply from 10.0.10.10: bytes=32 time=1ms TTL=127
Reply from 10.0.10.10: bytes=32 time<1ms TTL=127
Reply from 10.0.10.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.0.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.0.20.10
Pinging 10.0.20.10 with 32 bytes of data:
Request timed out.
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Reply from 10.0.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.0.20.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.0.30.10
Pinging 10.0.30.10 with 32 bytes of data:
Reply from 10.0.30.10: bytes=32 time<1ms TTL=127
Reply from 10.0.30.10: bytes=32 time<1ms TTL=127
Reply from 10.0.30.10: bytes=32 time<1ms TTL=127
Reply from 10.0.30.10: bytes=32 time<1ms TTL=127
Ping statistics for 10.0.30.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

9. At the Tel Aviv site, router-based DHCP was implemented, where each user VLAN is assigned a dedicated DHCP pool with the appropriate default gateway, eliminating the need for DHCP Relay.

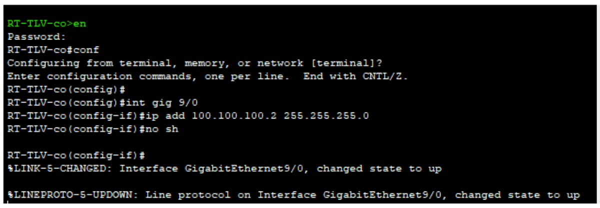
```
RT-TLV-co>en
Password:
RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp excluded-address 172.16.10.1 172.16.10.9
RT-TLV-co(config)#ip dhcp excluded-address 172.16.20.1 172.16.20.9
RT-TLV-co(config)#ip dhcp excluded-address 172.16.30.1 172.16.30.9
RT-TLV-co(config)#ip dhcp excluded-address 172.16.40.1 172.16.40.9
RT-TLV-co(config)#ip dhcp excluded-address 172.16.50.1 172.16.50.9
RT-TLV-co(config)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp pool it
RT-TLV-co(dhcp-config)#network 172.16.10.0 255.255.255.0
RT-TLV-co(dhcp-config)#default-router 172.16.10.1
RT-TLV-co(dhcp-config)#172.16.1.1
                        ^
% Invalid input detected at '^' marker.

RT-TLV-co(dhcp-config)#dns-server 172.16.1.1
RT-TLV-co(dhcp-config)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-TLV-co#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp pool pro
RT-TLV-co(dhcp-config)#network 172.16.20.0 255.255.255.0
RT-TLV-co(dhcp-config)#default-router 172.16.20.1
RT-TLV-co(dhcp-config)#dns-server 172.16.1.1
RT-TLV-co(dhcp-config)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp pool sel
RT-TLV-co(dhcp-config)#network 172.16.30.0 255.255.255.0
RT-TLV-co(dhcp-config)#default-router 172.16.30.1
RT-TLV-co(dhcp-config)#dns-server 172.16.1.1
RT-TLV-co(dhcp-config)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp pool mrk
RT-TLV-co(dhcp-config)#network 172.16.40.0 255.255.255.0
RT-TLV-co(dhcp-config)#default-router 172.16.40.1
RT-TLV-co(dhcp-config)#dns-server 172.16.1.1
RT-TLV-co(dhcp-config)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#ip dhcp pool gst
RT-TLV-co(dhcp-config)#network 172.16.50.0 255.255.255.0
RT-TLV-co(dhcp-config)#default-router 172.16.50.1
RT-TLV-co(dhcp-config)#dns-server 172.16.1.1
RT-TLV-co(dhcp-config)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console
```

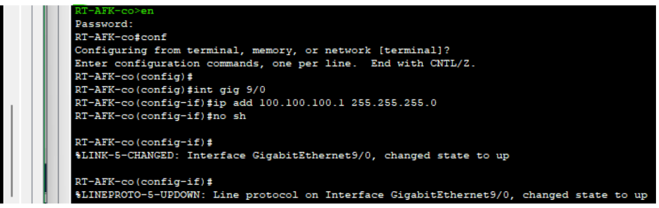
```
RT-TLV-co#sh ip dhcp pool

Pool it :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 5
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.10.1          172.16.10.1      - 172.16.10.254     2    / 5     / 254

Pool pro :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 5
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.20.1          172.16.20.1      - 172.16.20.254     0    / 5     / 254

Pool sel :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 5
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.30.1          172.16.30.1      - 172.16.30.254     0    / 5     / 254

Pool mrk :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 5
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.40.1          172.16.40.1      - 172.16.40.254     0    / 5     / 254

Pool gst :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 5
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 172.16.50.1          172.16.50.1      - 172.16.50.254     0    / 5     / 254
RT-TLV-co#
```

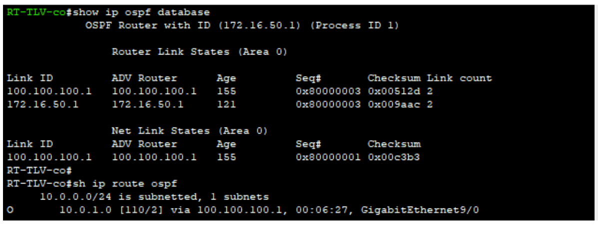
```
RT-TLV-co#sh ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.16.10.10     00D0.BA16.C658           --                     Automatic
172.16.10.11     0001.6461.9545           --                     Automatic
RT-TLV-co#
```

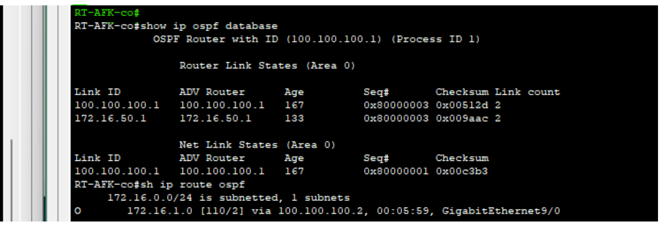


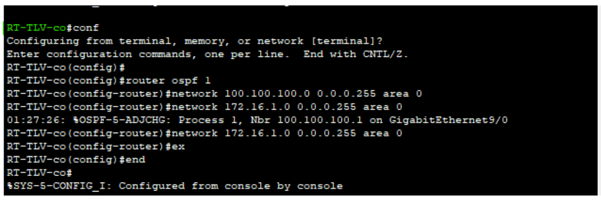
10. Dynamic routing between the company’s sites was implemented using OSPF Area 0, operating over a dedicated WAN link in the 100.100.100.0 network.

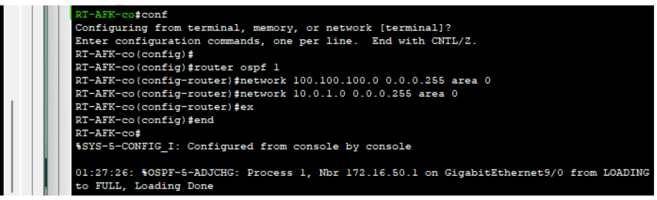




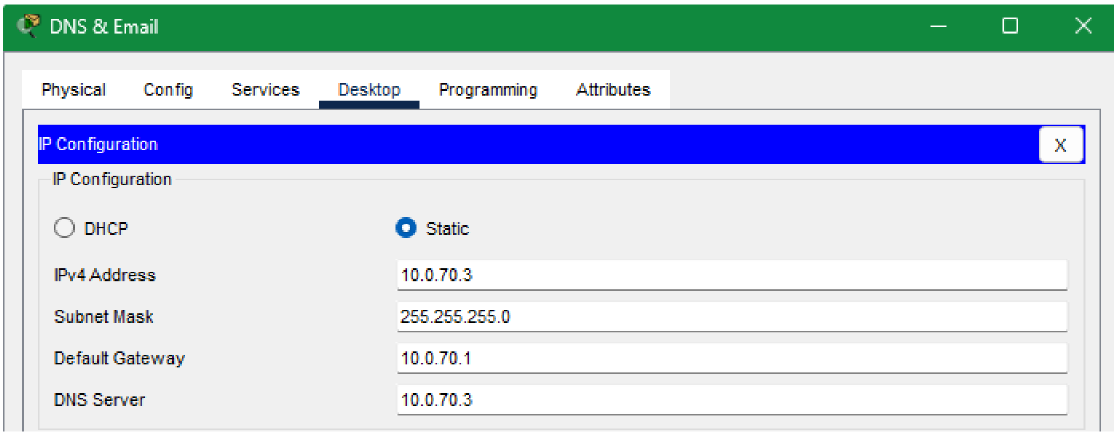


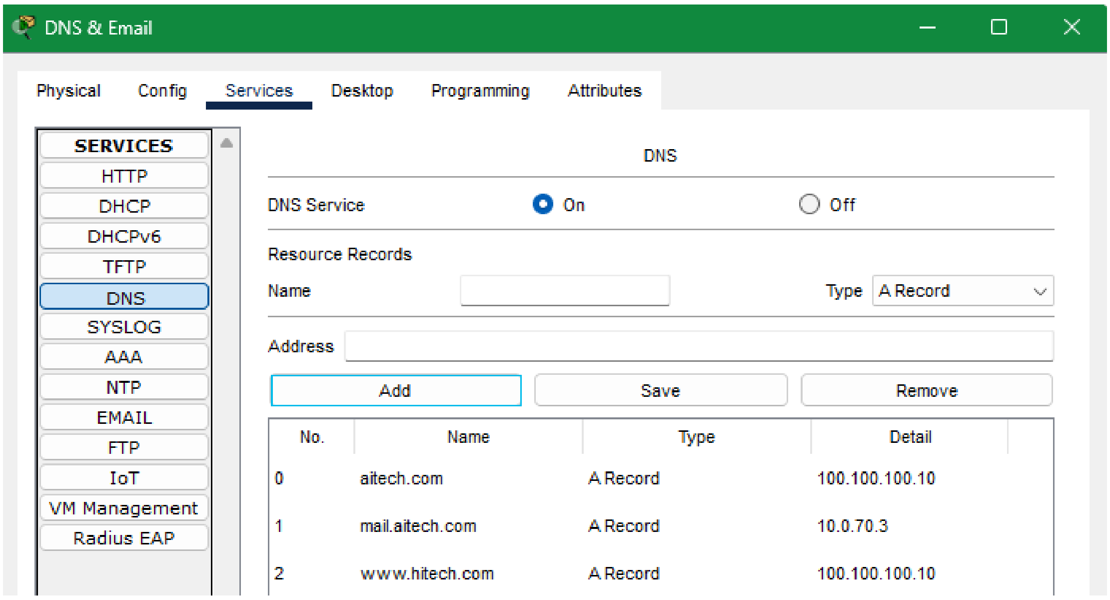


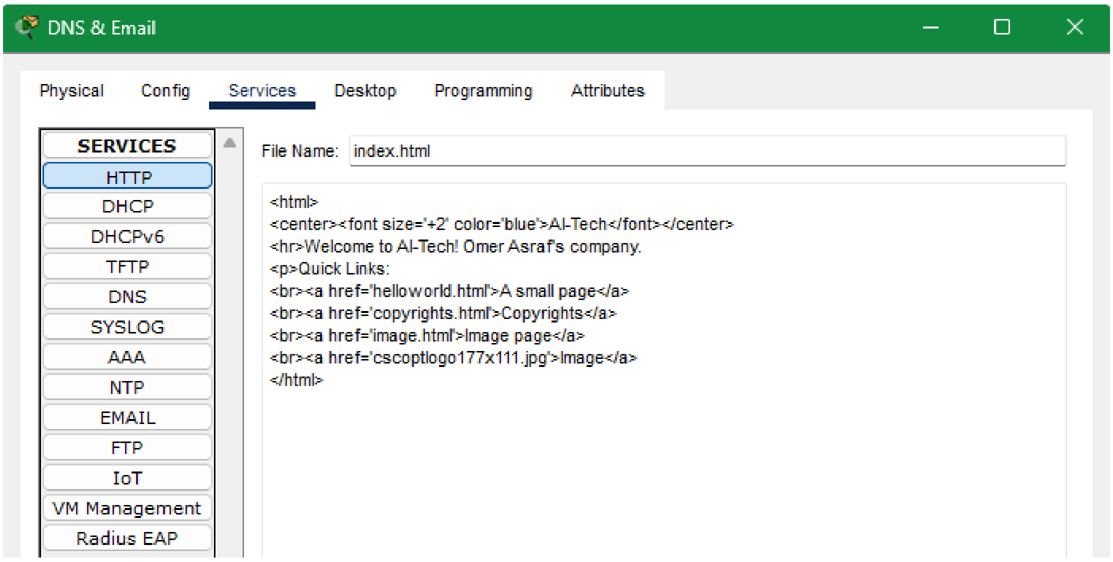


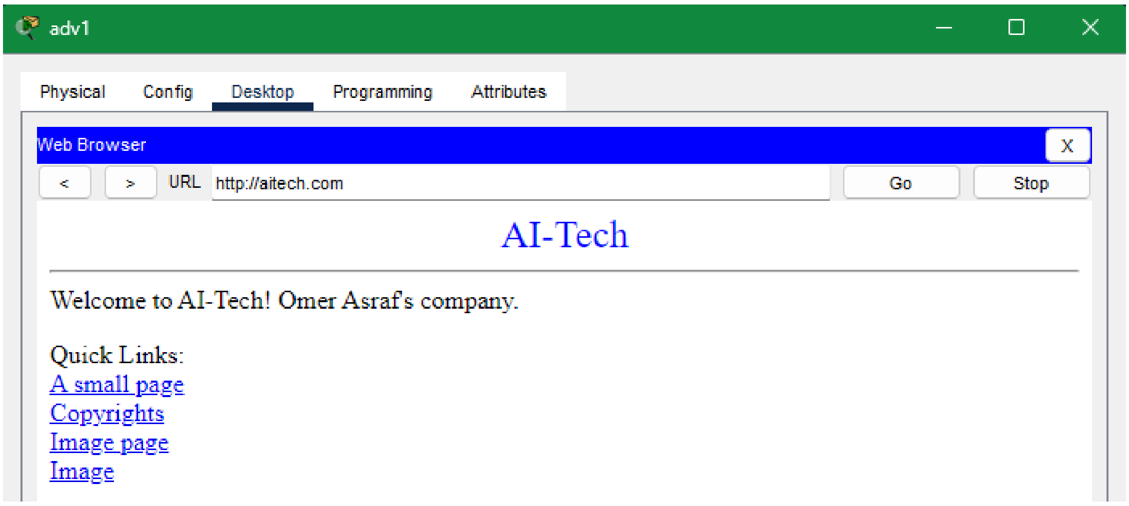


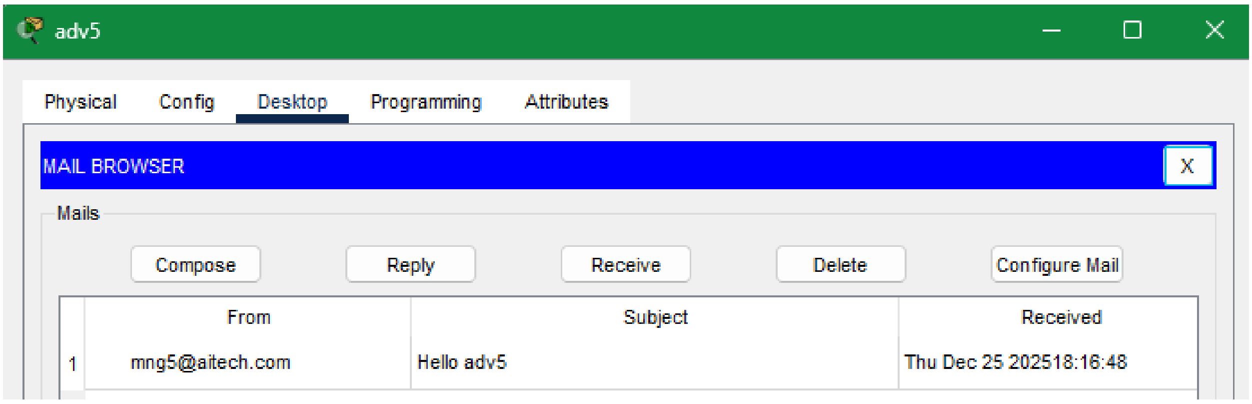
11. At the Afek site, a centralized server was deployed to provide DNS, Email, and Web services, where the DNS service maps the company domain to an external address for the corporate website and to an internal address for email services.



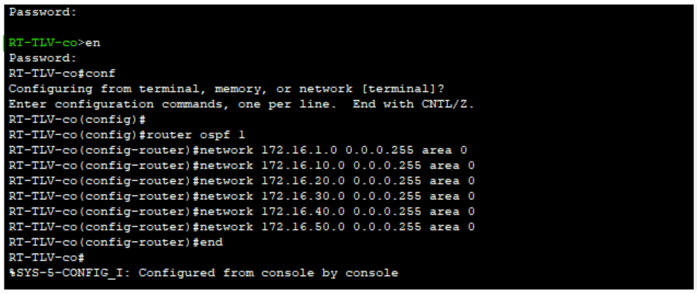


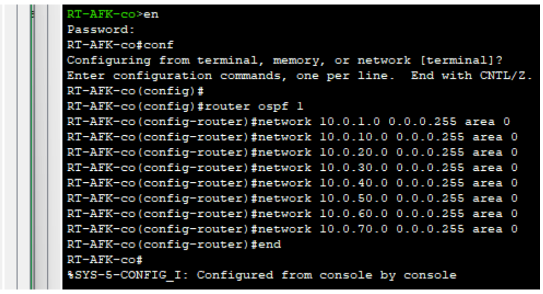






12. VLAN subnet propagation between sites was achieved through OSPF Area 0, where each VLAN-associated subnet was defined as a network statement on the core routers, ensuring full inter-site reachability.





Continue in next page

```
RT-TLV-co#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 8 subnets
O       10.0.1.0 [110/2] via 100.100.100.1, 00:06:43, GigabitEthernet9/0
O       10.0.10.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.20.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.30.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.40.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.50.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.60.0 [110/2] via 100.100.100.1, 00:01:35, GigabitEthernet9/0
O       10.0.70.0 [110/2] via 100.100.100.1, 00:01:22, GigabitEthernet9/0
     100.0.0.0/24 is subnetted, 1 subnets
C       100.100.100.0 is directly connected, GigabitEthernet9/0
     172.16.0.0/24 is subnetted, 6 subnets
C       172.16.1.0 is directly connected, GigabitEthernet8/0.111
C       172.16.10.0 is directly connected, GigabitEthernet8/0.10
C       172.16.20.0 is directly connected, GigabitEthernet8/0.20
C       172.16.30.0 is directly connected, GigabitEthernet8/0.30
C       172.16.40.0 is directly connected, GigabitEthernet8/0.40
C       172.16.50.0 is directly connected, GigabitEthernet8/0.50
```

it1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.50.10

Pinging 10.0.50.10 with 32 bytes of data:

Reply from 10.0.50.10: bytes=32 time<1ms TTL=126
Reply from 10.0.50.10: bytes=32 time<1ms TTL=126
Reply from 10.0.50.10: bytes=32 time<1ms TTL=126
Reply from 10.0.50.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ser1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.10.11

Pinging 172.16.10.11 with 32 bytes of data:

Reply from 172.16.10.11: bytes=32 time<1ms TTL=126
Reply from 172.16.10.11: bytes=32 time<1ms TTL=126
Reply from 172.16.10.11: bytes=32 time<1ms TTL=126
Reply from 172.16.10.11: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Section 13 in the next page

13. Access control was enforced using an Extended ACL applied to the guest interface at the Afek site, allowing users in VLAN 60 to access the company server only, while blocking access to all other internal departments.

```
RT-AFK-co>en
Password:
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#no ip access-list extended afk-gst-acl
RT-AFK-co(config)#^Z
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#ip access-list extended afk-gst-acl
RT-AFK-co(config-ext-nacl)#permit ip 10.0.60.0 0.0.0.255 host 10.0.70.3
RT-AFK-co(config-ext-nacl)#
RT-AFK-co(config-ext-nacl)#deny ip 10.0.60.0 0.0.0.255 10.0.0.0 0.255.255.255
RT-AFK-co(config-ext-nacl)#deny ip 10.0.60.0 0.0.0.255 172.16.0.0 0.255.255.255
RT-AFK-co(config-ext-nacl)#deny ip 10.0.60.0 0.0.0.255 10.0.1.0 0.0.0.255
RT-AFK-co(config-ext-nacl)#
RT-AFK-co(config-ext-nacl)#permit ip 10.0.60.0 0.0.0.255 any
RT-AFK-co(config-ext-nacl)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.60
RT-AFK-co(config-subif)#ip access-group afk-gst-acl in
RT-AFK-co(config-subif)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:FFFF:FE29:657D
   IPv6 Address....................: ::
   IPv4 Address....................: 10.0.60.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     10.0.60.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 10.0.70.3

Pinging 10.0.70.3 with 32 bytes of data:

Reply from 10.0.70.3: bytes=32 time<1ms TTL=127
Reply from 10.0.70.3: bytes=32 time<1ms TTL=127
Reply from 10.0.70.3: bytes=32 time<1ms TTL=127
Reply from 10.0.70.3: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.70.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.10.11

Pinging 10.0.10.11 with 32 bytes of data:

Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.

Ping statistics for 10.0.10.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.1.1

Pinging 10.0.1.1 with 32 bytes of data:

Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.
Reply from 10.0.60.1: Destination host unreachable.

Ping statistics for 10.0.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

14. At the Tel Aviv site, an Extended ACL was implemented to block guest users (VLAN 50) from accessing organizational email services, while preserving full email access for all other departments.

```
RT-TLV-co>en
Password:
RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip access-list extended tlv-gst-no-mail
RT-TLV-co(config-ext-nacl)#
RT-TLV-co(config-ext-nacl)#deny tcp 172.16.50.0 0.0.0.255 host 10.0.70.3 eq 25
RT-TLV-co(config-ext-nacl)#deny tcp 172.16.50.0 0.0.0.255 host 10.0.70.3 eq 110
RT-TLV-co(config-ext-nacl)#
RT-TLV-co(config-ext-nacl)#permit ip 172.16.50.0 0.0.0.255 any
RT-TLV-co(config-ext-nacl)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.50
RT-TLV-co(config-subif)#ip access-group tlv-gst-no-mail in
RT-TLV-co(config-subif)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console
```

Section 15 in the next page

15. A security policy was enforced using an Extended ACL that restricts ICMP initiation, allowing only the IT department (VLAN 10) at the Tel Aviv site to perform ping operations, while all other departments across the organization are prevented from initiating ICMP traffic without impacting other network services.

```
RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#ip access-list extended tlv-no-icmp-echo
RT-TLV-co(config-ext-nacl)#
RT-TLV-co(config-ext-nacl)#deny icmp any any echo
RT-TLV-co(config-ext-nacl)#
RT-TLV-co(config-ext-nacl)#permit icmp any any echo-reply
RT-TLV-co(config-ext-nacl)#permit ip any any
RT-TLV-co(config-ext-nacl)#
RT-TLV-co(config-ext-nacl)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-TLV-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.20
RT-TLV-co(config-subif)#ip access-group tlv-no-icmp-echo in
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.30
RT-TLV-co(config-subif)#ip access-group tlv-no-icmp-echo in
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.40
RT-TLV-co(config-subif)#ip access-group tlv-no-icmp-echo in
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#int gig 8/0.111
RT-TLV-co(config-subif)#ip access-group tlv-no-icmp-echo in
RT-TLV-co(config-subif)#ex
RT-TLV-co(config)#
RT-TLV-co(config)#end
RT-TLV-co#
%SYS-5-CONFIG_I: Configured from console by console
```

```
RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#ip access-list extended afk-no-icmp-echo
RT-AFK-co(config-ext-nacl)#
RT-AFK-co(config-ext-nacl)#deny icmp any any echo
RT-AFK-co(config-ext-nacl)#
RT-AFK-co(config-ext-nacl)#permit icmp any any echo-reply
RT-AFK-co(config-ext-nacl)#permit ip any any
RT-AFK-co(config-ext-nacl)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console

RT-AFK-co#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.10
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.20
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.30
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.40
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.50
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#
RT-AFK-co(config)#int gig 8/0.60
RT-AFK-co(config-subif)#ip access-group afk-no-icmp-echo in
RT-AFK-co(config-subif)#ex
RT-AFK-co(config)#end
RT-AFK-co#
%SYS-5-CONFIG_I: Configured from console by console
```

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:BAFF:FE16:C658
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.10.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     172.16.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 172.16.30.10

Pinging 172.16.30.10 with 32 bytes of data:

Reply from 172.16.30.10: bytes=32 time<1ms TTL=127
Reply from 172.16.30.10: bytes=32 time<1ms TTL=127
Reply from 172.16.30.10: bytes=32 time<1ms TTL=127
Reply from 172.16.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>
C:\>
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::207:ECFF:FE22:353C
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.40.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     172.16.40.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 172.16.30.10

Pinging 172.16.30.10 with 32 bytes of data:

Reply from 172.16.40.1: Destination host unreachable.
Reply from 172.16.40.1: Destination host unreachable.
Reply from 172.16.40.1: Destination host unreachable.
Reply from 172.16.40.1: Destination host unreachable.

Ping statistics for 172.16.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Summary

This project demonstrated the complete design, implementation, and validation of a multi-site enterprise network infrastructure in accordance with industry best practices.

The network was built using a hierarchical architecture, incorporating logical segmentation through VLANs, dynamic routing for inter-site connectivity, centralized network services, and secure device management.

The project highlights how structured planning, proper technology selection, and policy-based security controls enable the creation of a scalable, secure, and manageable enterprise network that meets both operational and security requirements.